

6.76e)

$$(2-j) + 4 - 3(2+j) - j$$

$$2-j+4-6-3j-j = \underline{\underline{-5j}} \quad \checkmark$$

6.20e)

$$(1-3j)(2+3j)j = (2+3j-6j-9j^2)j =$$

$$= (2-3j+9)j = (11-3j)j =$$

$$= 11j - 3j^2 = \underline{\underline{3+11j}} \quad \checkmark$$

6.22g)

$$\frac{(1+2j)(3-j)}{(-2+j)(1+j)} = \frac{3-j+6j-2j^2}{-2-2j+j+j^2} = \frac{5+5j}{-3-j} \frac{(-3+j)}{(-3+j)} =$$

$$= \frac{-15+5j-15j+5j^2}{10} = \frac{-20-10j}{10} = \underline{\underline{-2-j}} \quad \checkmark$$